

2.4 Squares and square roots

You will learn to:

- Use index notation for squares and square roots.
- Calculate with squares and square roots.

$$E = mc^2$$

Confidence

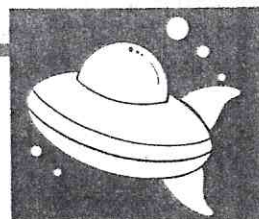
Why learn this?

Square numbers are used to work out the trajectory of a football, or the equivalence of mass and energy, or the area you need to tile.

Fluency

Work out

- 5×5
- 8×8
- 0.3×3
- 0.3×0.3
- 0.5×5
- 0.5×0.5



Explore

What is the best way to arrange 8000 s into square blocks for a music concert?

Exercise 2.4

Warm up

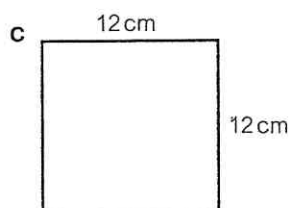
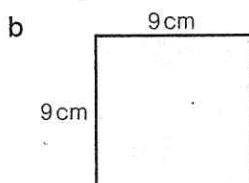
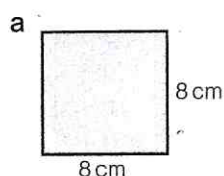
- 1 You can arrange 9 dots in a square like this.



Which numbers from this list can be arranged as dots in a square?

2 4 8 12 16 20 24 36 42 49 55

- 2 Work out the area of each square.



Discussion How do you find the area of a square using the length of one side?

- 3 a Copy and complete this table of square numbers from 1 to 15.

1^2	2^2	3^2	4^2	5^2	6^2	7^2	8^2	9^2	10^2	11^2	12^2	13^2	14^2	15^2
1	4													

- b Use your table in part a to write down the answers to these.

i $\sqrt{64}$

ii $\sqrt{121}$

iii $\sqrt{169}$

iv $\sqrt{225}$

Discussion Is there another **square root** of 64, 121, 169, 225?

Key point

To find the square of a number you multiply it by itself.

$$3 \times 3 = 3^2 = 9$$

3^2 means '3 squared'.

Key point

The inverse of square is square root.

$$3^2 = 3 \times 3 = 9, \text{ so a square root of } 9 = \sqrt{9} = 3$$

Check each of these calculations is correct by using the inverse operation.

- a $14^2 = 196$
- b $2.5^2 = 6.25$
- c $\sqrt{2116} = 46$
- d $\sqrt{12.96} = 3.6$

Q4 hint

- a Work out $\sqrt{196}$.
- c Work out 46^2 .

Worked example

Work out an estimate of $\sqrt{55}$.

$\sqrt{49} = 7$ and $\sqrt{64} = 8$

$\sqrt{55}$ lies between 7 and 8

Estimate is 7.4

55 is between 49 and 64

55 is closer to 49 than 64, so estimate just less than 7.5



5 a Work out an estimate for these square roots.

- i $\sqrt{20}$ ii $\sqrt{5}$ iii $\sqrt{79}$ iv $\sqrt{90}$

b Check your estimates by working out the accurate square roots on a calculator.

6 a Work out

- i 2^2 ii 20^2 iii 200^2 iv 0.2^2

Discussion What do you notice about your answers to part a?

b Work out

- i 50^2 ii 900^2 iii 4000^2 iv 0.6^2

7 Work out

- a $\sqrt{36}$ b $\sqrt{9} \times \sqrt{4}$ c $\sqrt{324}$ d $\sqrt{784}$

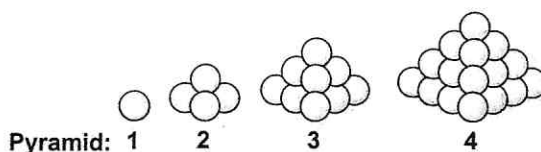
Discussion What do you notice about your answers to parts a and b?

Q5a hint

Use your table from Q3 to help you.

Investigation

The diagram shows four pyramids made from balls.



1 Copy and complete this table.

Pyramid	1	2	3	4
Number of balls	$1^2 = 1$	$1^2 + 2^2 = \square$		

2 How many balls will be in

- a pyramid 5 b pyramid 6?

3 Explain how you can work out the number of balls in pyramid 12 without working out the number of balls in pyramids 1 to 11 first.

Reasoning

8 **Explore** What is the best way to arrange 8000 seats into square blocks for a music concert?

Is it easier to explore this question now you have completed the lesson?
What further information do you need to be able to answer this?

9 **Reflect** Think about the *square* of 100 and the *square root* of 100. Which is 10 and which is 10000? Make sure you know the difference between these two terms.

Write down a definition in your own words to help you remember.



Explore

Reflect

Squares and square roots

- 1 a Complete this table showing all the square numbers from 1 to 15.

1^2	2^2	3^2	4^2	5^2	6^2	7^2	8^2	9^2	10^2	11^2	12^2	13^2	14^2	15^2
1	4													

- b Use your table in part a to write down the answers to these.

i $\sqrt{9}$

ii $\sqrt{49}$

iii $\sqrt{144}$

To find the square of a number you multiply it by itself.
 $3^2 = 3 \times 3 = 9$
 3^2 means '3 squared'.

- 2 Reasoning A square has area 64 cm^2 .

Write down the side length of the square.

Onick says, 'The side length of the square could be 8 cm or -8 cm because the square root of 64 is 8 or -8.'

Tammy says, 'The side length of the square can only be 8 cm.'

Who is correct? Explain your answer.



- 3 Check each of these calculations is correct by using the inverse operation.

a $19^2 = 361$

b $5.5^2 = 30.25$

c $\sqrt{24.01} = 4.9$

- a Work out $\sqrt{361}$.
 c Work out 4.9^2 .

Guided

- 4 a Work out an estimate for these square roots.

i $\sqrt{28}$ $\sqrt{25} = 5$, $\sqrt{36} = 6$. So $\sqrt{28}$ lies between 5 and 6.
 Estimate is 5.3

ii $\sqrt{47}$

iii $\sqrt{110}$

Use your table from Q1 to help you.
 28 is closer to 25 than 36, so estimate a bit less than 5.5.



- b Check your estimates by working out the accurate square roots on a calculator.

- 5 Work out

a 3^2

b 30^2

c 300^2

d 0.3^2

The inverse of square is **square root**.
 $3^2 = 3 \times 3 = 9$, so the square root of 9 is $\sqrt{9} = 3$

- 6 Work out

a $\sqrt{400}$

c $\sqrt{900}$

e $\sqrt{8100}$

b $\sqrt{4} \times \sqrt{100}$

d $\sqrt{9} \times \sqrt{100}$

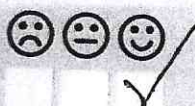
f $\sqrt{3969}$



$49 \times 81 = 3969$

Check

Tick each box as your confidence in this topic improves.



Need extra help? Go to page 18 and tick the boxes next to Q8 and 9. Then try them once you've finished 2 1-2 6

2.5 More power and roots

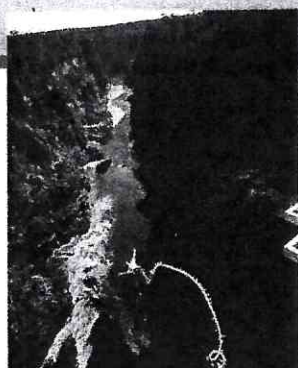
You will learn to:

- Carry out calculations involving squares, cubes, square roots and cube roots.
- Use factorising to work out square roots and cube roots.
- Solve word problems using square roots and cube roots.

Confidence

Why learn this?

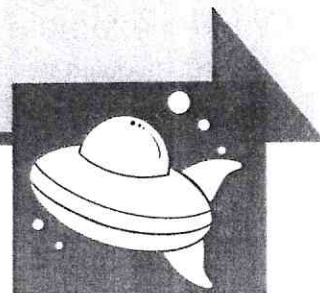
Scientists can use square roots to estimate how long an object takes to fall.



Fluency

Work out

- 5^2
- $\sqrt{36}$



Explore

For how long are you falling in a bungee jump (jump from an elasticated rope) from the Sidu River Bridge in China?

Exercise 2.5

1 Work out

a $5 \times 5 \times 5$

b $3 \times 3 \times 3$

2 Work out

a $7 + 5 \times 9$

b $14 - \frac{8}{2}$

c $3 \times 4 + 7 \times 5$

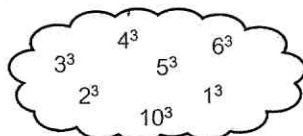
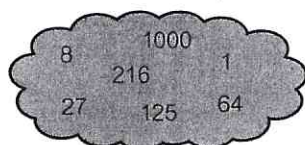
3 Write these calculations using **index notation**.

a $3 \times 3 \times 3 \times 3 = 3^{\square}$

b $2 \times 2 \times 2 \times 2 \times 2$

c $7 \times 7 \times 7$

4 Match each **cube** number in the cloud to a number written in index notation in the other cloud.



5 Work out these **powers** of 10.

a 10^3

b 10^5

c 10^4

d 10^6

6 Work out these **cube roots**.

a $\sqrt[3]{27}$

b $\sqrt[3]{125}$

c $\sqrt[3]{1000}$

d $\sqrt[3]{1}$

e $\sqrt[3]{-8}$

f $\sqrt[3]{-64}$

7 Work out the missing numbers.

a $\sqrt[3]{216} = \sqrt[3]{8} \times \sqrt[3]{27} = 2 \times \square = \square$

b $\sqrt[3]{8000} = \sqrt[3]{64} \times \sqrt[3]{125} = \square \times \square = \square$

c $\sqrt[3]{1728} = \sqrt[3]{27} \times \sqrt[3]{64} = \square \times \square = \square$

Key point

You can use **index notation** to write a number to a **power** or **index**.

The power tells you how many times the number is multiplied by itself.

$2^3 = 2 \times 2 \times 2$

2^3 is '2 to the power 3'.

3 is the power.

Key point

To find the **cube** of a number, multiply it by itself and then multiply by the number again.

2 cubed $= 2 \times 2 \times 2$

Key point

The inverse of cube is **cube root**.

$2^3 = 8$, so the cube root of 8 is $\sqrt[3]{8} =$

Q6e hint

$\square \times \square \times \square = 8$

$\square \times \square \times \square = -8$

8 Work out

a $12 - 12$

d $\frac{63}{4} - 15$

b $3 + 9^2$

e $\sqrt{49} + 5 \times 8$

c $4 \times 3^2 + 5^2$

f $\frac{\sqrt{100}}{2} + 10^2$

9 Work out

a 3×2^3

b $10^3 \times 5$

c $\frac{10^3}{500}$

d $\frac{40}{2^3}$

e $4 \times \sqrt[3]{64}$

f $\frac{24}{\sqrt[3]{8}}$

10 Work out

a $50 - 3^3$

b $4^3 + 6^2$

c $6 \times 2^3 - 18$

d $4 \times \sqrt[3]{1000} - 35$

e $\frac{20}{\sqrt[3]{125}} - 2^2$

11 Use a calculator to work out

a $7^3 + 27$

b $2 \times 8^3 - 624$

c $6 \times 10^3 - 4 \times 9^3$

d $4 \times \sqrt[3]{729} + 4$

e $\frac{\sqrt[3]{1728}}{6} - 2$

f $4 \times \sqrt[3]{216} - 5^2$

Key point



Squares, cubes, square roots are indices or powers.

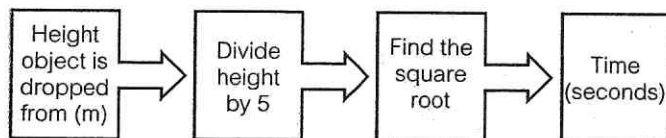
The priority of operations is

- Brackets
- Indices or Powers
- Multiplication and Division
- Addition and Subtraction

Q11 hint

Use the $\sqrt[3]{}$ and x^y buttons on your calculator.

12 **STEM / Modelling** You can estimate the time it will take an object to fall using this flowchart.



Work out the time it will take a ball to drop from these heights.

a 80 m

b 125 m

c 405 m

13 **Problem-solving / Reasoning** One number is missing from this list.

$\sqrt{49} \times 40$ 30^2 10×5^2 $\frac{800^2}{1000}$ $\square$

The range of the numbers is 800.

a Work out the missing number.

b Is there only one possible answer to part a? Explain why.

14 Work out

a $\sqrt[3]{8}$

b $8\sqrt[3]{27}$

c $8\sqrt[3]{8 \times 27}$

What do you notice?

15 **Problem-solving** Work out $\sqrt[3]{512}$.

16 **Problem-solving** $13824 = 8 \times 27 \times 64$

Use this fact to work out $\sqrt[3]{13824}$.

17 **Explore** For how long are you falling in a bungee jump from the Sidu River Bridge in China?

Is it easier to explore this question now you have completed the lesson?

What further information do you need to be able to answer this?

18 **Reflect** The $\sqrt{}$ root symbol began as the letter r in the 16th Century. You could remember r for root! List the mathematics notation used in this lesson, and ways you might remember it.

Q15 hint

Work out which two cube numbers multiply to give 512.

Q18 hint

'Notation' means the signs and symbols you use.

More powers and roots

- 1 Complete this table showing commonly used cube numbers.

1^3	2^3	3^3	4^3	5^3	6^3	10^3
1	8				216	

To find the cube of a number, multiply it by itself and then multiply by the number again.
 $2^3 = 2 \times 2 \times 2 = 8$
 2^3 means '2 cubed' or '2 to the power 3'.
 3 is the power.

The inverse of cube is **cube root**.

$2^3 = 8$, so the cube root of 8 is $\sqrt[3]{8} = 2$.

- 2 Use your table in Q1 to write down the answers to these.

a $\sqrt[3]{8}$

b $\sqrt[3]{216}$

d $\square \times \square \times \square = 27$

c $\sqrt[3]{64}$

d $\sqrt[3]{-27}$

$\square \times \square \times \square = -27$

- 3 Work out

a $5^2 + 5 = 25 + 5 =$ b $2 \times 4^2 + 1 = 32 + 1$

c $\frac{8^2}{2}$ d $\frac{\sqrt{81}}{3} + 3 \times 9$

Squares, cubes, square roots are indices or powers.

The priority of operations is

- Brackets
- Indices or Powers
- Multiplication and Division
- Addition and Subtraction

- 4 Work out

a $2^3 \times 5$ b 10×3^3

c $\frac{5^3}{25}$ d $5 \times \sqrt[3]{1000}$

e $\frac{50}{\sqrt[3]{125}}$ f $6^2 - 3^3$

g $2 \times \sqrt[3]{64} + 2$ h $\frac{4^3}{\sqrt[3]{64}} - 2$

- 5 Use a calculator to work out

a $8^3 - 12$

b $3 \times 21^2 - 323$

c $31 \times 12^2 - 0.5 \times 12^3$

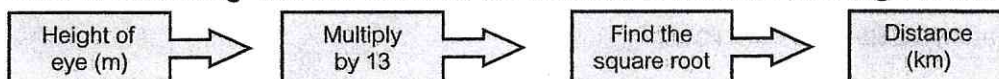
d $5 \times \sqrt[3]{512} + 10$

e $\frac{\sqrt[3]{2744}}{\sqrt{784}} + 4.5$



Use the $\sqrt[3]{}$ and x^y buttons on your calculator.

- 6 **STEM / Modelling** You can estimate the distance to the horizon using this flowchart.



Work out the distance to the horizon when it is seen from a height of

a 1.5m

b 10m

c 20m

Check

Tick each box as your **confidence** in this topic improves.



Need extra help? Go to page 18 and tick the boxes next to Q10 and 11. Then try them once you've finished 2.1–2.6.